Supplementary Material

Mathematical derivations and computational construction of the figures

Beyond the Classical Parabola: Tafelmacher's Construction in Proportional Arithmetic

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Abstract

This supplement documents the mathematical definitions, parametrisations, computational workflow, and tangency checks used to construct Figures 1–4. The classical construction and its proportional-arithmetic counterpart are defined separately before the logarithmic correspondence is used for verification. A supplementary eight-tangent visualisation is also included.

1 Arithmetic conventions and plotting coordinates

The ordinary geometric coordinates are $(x, y) \in \mathbb{R}^2$. Proportional coordinates $(X, Y) \in (0, \infty)^2$ are related by

$$(X, Y) = (e^x, e^y), \qquad (x, y) = (\ln X, \ln Y). \tag{1}$$

The proportional operations on positive numbers are

$$X \oplus Y = XY, \qquad X \ominus Y = X/Y, \tag{2}$$

$$X \odot Y = \exp((\ln X)(\ln Y)), \qquad X \oslash Y = \exp\left(\frac{\ln X}{\ln Y}\right), \quad Y \neq 1. \tag{3}$$

Write $\mathcal{E}(r) = e^r$. The proportional zero is 1, and the proportional multiplicative unit is e. A proportional square means $X^{\{2\}} = X \odot X$, *not* the ordinary square X^2. The internally defined interpolation is

$$I_t(U, V) = [\mathcal{E}(1-t) \odot U] \oplus [\mathcal{E}(t) \odot V] = U^{1-t}V^t, \quad 0 \le t \le 1. \tag{4}$$

The right-hand expression is an ordinary-coordinate display identity; the left-hand expression specifies the operation within proportional arithmetic.

Unless otherwise indicated, the figures use $a = h = 1$ and ordinary linear plotting axes, including the panels displaying proportional coordinates. Thus proportional straight lines need not look straight on the page.

2 Figure 1: equal divisions in two arithmetics

For an ordinary interval with endpoints u, v, the $n+1$ equally spaced points are

$$u_k = u + \frac{k}{n}(v-u), \quad k = 0, \dots, n, \qquad u_{k+1} - u_k = \frac{v-u}{n}. \tag{5}$$

For positive proportional endpoints U, V, define instead

$$U_k = I_{k/n}(U, V) = U^{1-k/n}V^{k/n}, \qquad \frac{U_{k+1}}{U_k} = \left(\frac{V}{U}\right)^{1/n}. \tag{6}$$

The ordinary increments are equal in the first panel; the ratios are equal in the second. To plot the points, compute the two arrays and mark their positions on linear axes. The distinction is between equal differences and equal ratios, not between two different axis scales.

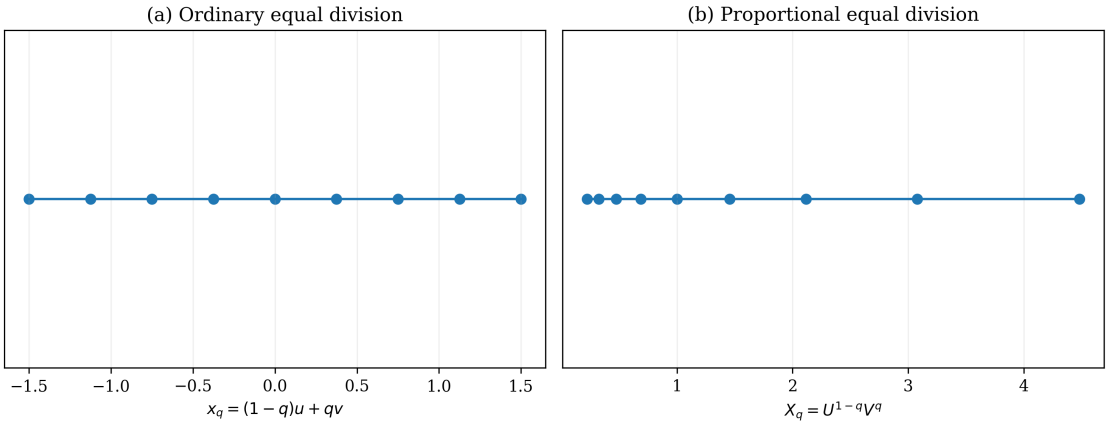


Figure 1: Figure 1: ordinary equal differences versus proportional equal ratios.

3 Figure 2: Tafelmacher's joining-line construction

Choose the classical triangle

$$A = (0, h), \quad B = (-a, 0), \quad C = (a, 0), \qquad a, h > 0. \tag{7}$$

Use complementary divisions of the sides:

$$P(t) = (1-t)A + tB = (-at, h(1-t)), \tag{8}$$
$$Q(t) = tA + (1-t)C = (a(1-t), ht). \tag{9}$$

Their joining line has equation

$$F(x, y, t) := h(1-2t)x + ay - ah(1-2t+2t^2) = 0. \tag{10}$$

For the discrete construction, use $t_q = q/(2n)$ with $q = 1, \ldots, 2n-1$. In the accompanying Figure 2 Python script, $n = 8$, so there are fifteen interior members of the family.

The proportional vertices are

$$A_p = (1, e^h), \quad B_p = (e^{-a}, 1), \quad C_p = (e^a, 1).$$

Using (4) componentwise gives

$$P_p(t) = I_t(A_p, B_p) = (e^{-at}, e^{h(1-t)}), \tag{11}$$
$$Q_p(t) = I_{1-t}(A_p, C_p) = (e^{a(1-t)}, e^{ht}). \tag{12}$$

The proportional joining line is defined by

$$[\mathcal{E}(h(1-2t)) \odot X] \oplus [\mathcal{E}(a) \odot Y] = \mathcal{E}(ah(1-2t+2t^2)). \tag{13}$$

For plotting on ordinary (X, Y) axes, solve (13) for Y to obtain the power curve

$$Y_t(X) = \exp\big(h(1-2t+2t^2)\big) X^{(h/a)(2t-1)}, \qquad X > 0. \tag{14}$$

Each proportional side is plotted using sampled proportional interpolation, rather than an ordinary Euclidean segment.

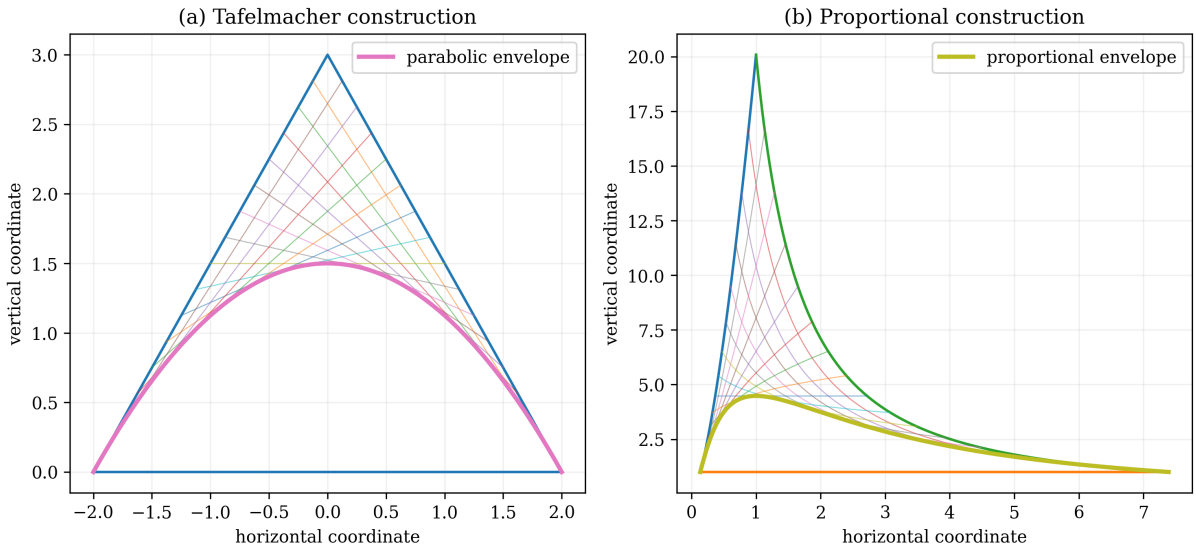


Figure 2: Figure 2: discrete Tafelmacher construction with classical joining lines and proportional joining power curves.

4 Figure 3: the two parabola loci

The classical envelope is

$$y = f(x) = \frac{h}{2}\left(1 - \frac{x^2}{a^2}\right), \quad -a \le x \le a. \tag{15}$$

The proportional parabola is first defined using proportional arithmetic:

$$Y = (H_p \oslash \mathcal{E}(2)) \odot \left[e \ominus \left(X^{\{2\}} \oslash W^{\{2\}}\right)\right], \quad H_p = e^h,\ W = e^a. \tag{16}$$

Applying the logarithm to (16) yields the plotting identity

$$Y = g(X) = \exp\left[\frac{h}{2}\left(1 - \frac{(\ln X)^2}{a^2}\right)\right], \quad e^{-a} \le X \le e^a. \tag{17}$$

In particular, for $a = h = 1$, the proportional parabola has a maximum at $(X, Y) = (1, \sqrt{e})$, and its two endpoint values are $Y = 1$. It is generally asymmetric on ordinary linear axes, although its logarithmic-coordinate counterpart is a symmetric parabola.

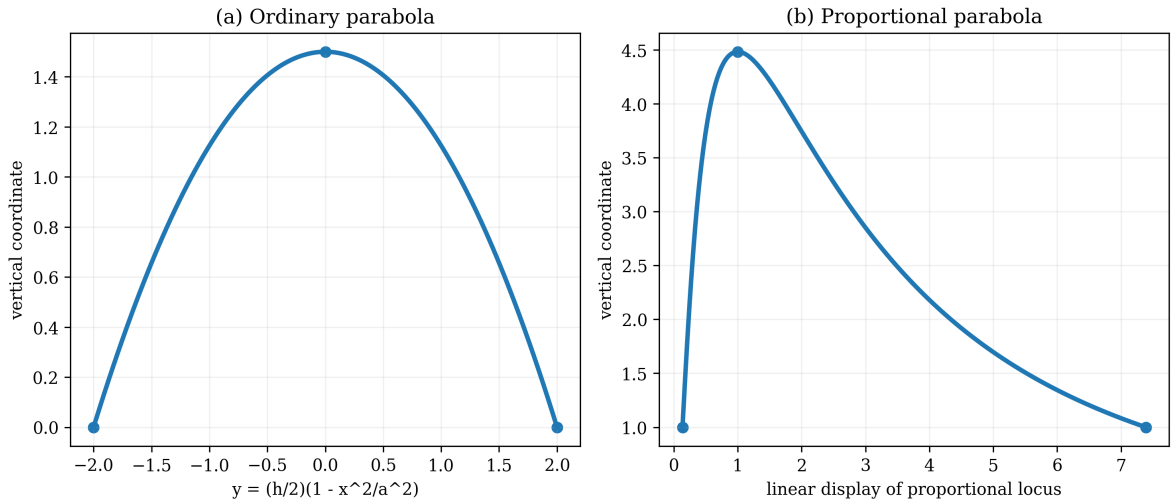


Figure 3: Figure 3: the classical parabola and the internally defined proportional parabola shown on ordinary linear axes.

5 Figure 4: exact tangency and contact points

To locate the envelope of (10), solve $F(x, y, t) = 0$ and $F_t(x, y, t) = 0$. Since

$$F_t = -2hx + 2ah - 4aht,$$

the contact point is

$$T(t) = (a(1 - 2t),\ 2ht(1 - t)). \tag{18}$$

The derivative of (15) at the contact point is

$$f'(a(1 - 2t)) = -\frac{h}{a}(1 - 2t) = \frac{h}{a}(2t - 1),$$

equal to the slope of (10). Thus the line is tangent.

In proportional coordinates the contact point is

$$T_p(t) = (e^{a(1-2t)}, e^{2ht(1-t)}). \tag{19}$$

The proportional tangent is the power curve (14). Its derivative is

$$Y_t'(X) = \frac{h}{a}(2t - 1)\frac{Y_t(X)}{X}.$$

The derivative of the proportional parabola is

$$g'(X) = -\frac{h \ln X}{a^2 X}\, g(X). \tag{20}$$

At $X = e^{a(1-2t)}$, both derivatives equal $\frac{h}{a}(2t - 1)\frac{e^{2ht(1-t)}}{e^{a(1-2t)}}$. Both curves also pass through $T_p(t)$; hence tangency is exact.

The four-tangent version uses $t \in \{0.18, 0.38, 0.62, 0.82\}$. Each point is computed from (18) or (19), not estimated from intersections of plotted curves.

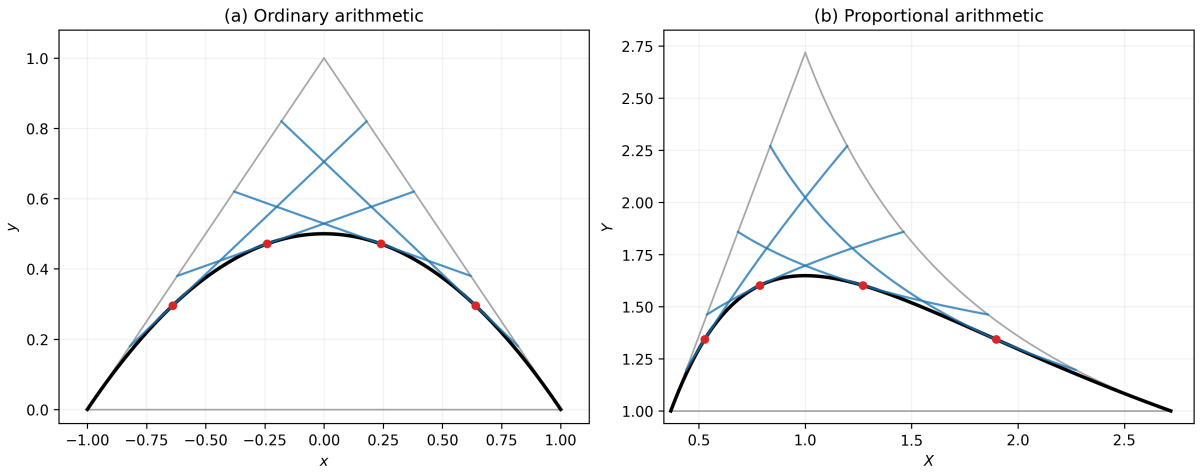


Figure 4: Figure 4: four exact tangencies in classical and proportional arithmetic, with analytically computed contact points.

6 Supplementary extension: eight proportional tangents

For the extended illustration, retain the previous four values and add $t \in \{0.10, 0.28, 0.72, 0.90\}$. The complete ordered list is

$$t \in \{0.10, 0.18, 0.28, 0.38, 0.62, 0.72, 0.82, 0.90\}.$$

With $a = h = 1$, the eight power curves have the explicit form

$$Y_t(X) = e^{1-2t+2t^2} X^{2t-1}, \tag{21}$$

and their contact points are

$$(X_t, Y_t) = \left(e^{1-2t}, e^{2t(1-t)}\right).$$

The displayed black locus is $Y = \exp((1 - (\ln X)^2)/2)$.

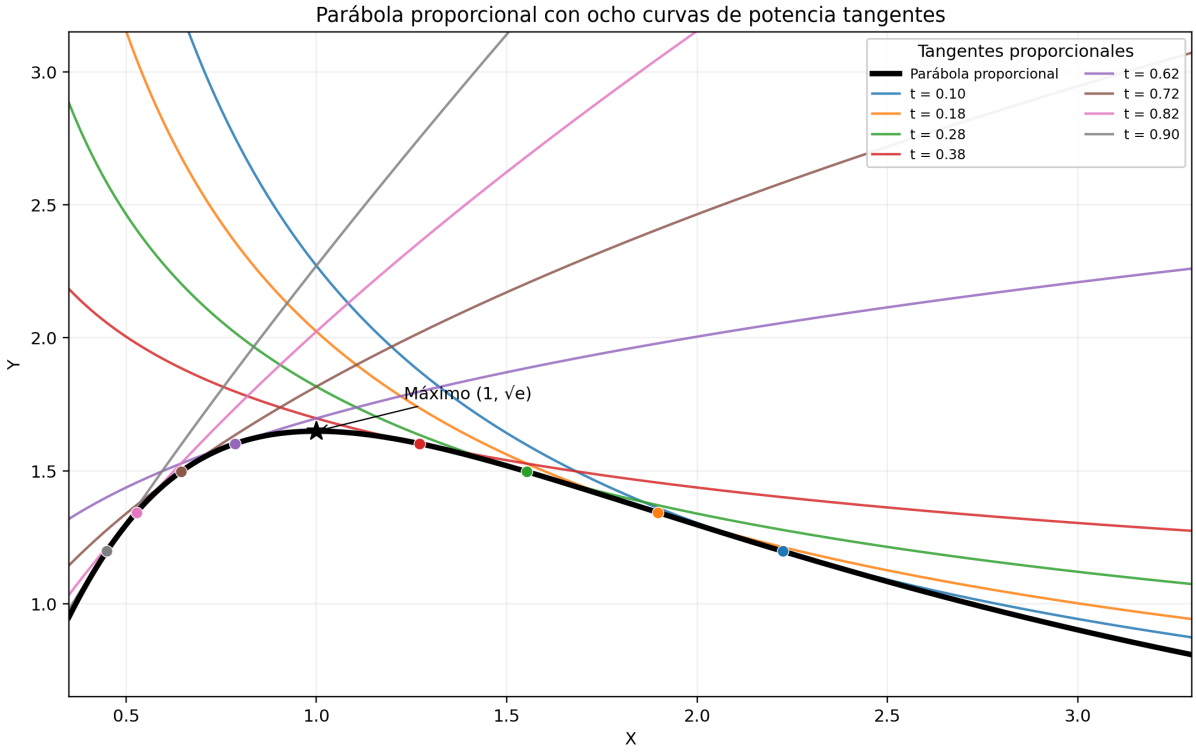


Figure 5: Supplementary Figure S1: eight power-curve tangents to the proportional parabola. Coloured markers indicate exact contacts.

7 Reproducible numerical workflow

The computations use Python, NumPy and Matplotlib. The following pseudocode specifies the essential numerical steps:

```
a = h = 1.0
t_values = [0.18, 0.38, 0.62, 0.82]
x = linspace(-a, a, 600)
X = linspace(exp(-a), exp(a), 600)
ordinary_parabola = h/2 * (1 - (x/a)**2)
proportional_parabola = exp(h/2 * (1-(log(X)/a)**2))
for t in t_values:
    ordinary_tangent = h/a*(2*t-1)*x + h*(1-2*t+2*t*t)
    power_tangent = exp(h*(1-2*t+2*t*t))*X**(h/a*(2*t-1))
    ordinary_contact = (a*(1-2*t), 2*h*t*(1-t))
    proportional_contact = exp(ordinary_contact)
```

The implementation used to generate the original Figure 2 and Figure 4 also defines the proportional operations explicitly, including the internal proportional parabola (16). For numerical verification, evaluate the residuals of (10), (15), (13) and (16) at the analytically defined points; residuals should be close to zero within floating-point tolerance. The symbolic identities above provide the proof.

8 Interpretation and limitations

A proportional line is a line with respect to the transported arithmetic structure; it is usually a power curve when drawn on ordinary Cartesian axes. The tangency result follows from the logarithmic field isomorphism and can also be checked using ordinary derivatives of the display curves.

It should not be interpreted as an independent new Euclidean incidence theorem. The primary purpose of the visualisations is to make the arithmetic dependence of a geometric construction explicit.

References

[1] A. Tafelmacher, "De la parábola," *Anales del Instituto de Ingenieros de Chile*, year III, no. 5, pp. 187–192, 15 May 1903.